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MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026
Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEBCA302 - MACHINE LEARNING

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Very Short Answer Questions
Answer all questions
Each question carries 2 marks

1. **What is a training dataset? What is its purpose in Machine Learning?** [K] / [CO1]
   A training dataset is a collection of data used to train a Machine Learning model by helping it learn patterns and relationships in the data. - **1 mark**. Purpose: It enables the model to learn from examples and make predictions or decisions on new, unseen data. – **1 mark**
2. **What is Machine Learning?** [K] / [CO1]
   Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn from data and improve their performance without being explicitly programmed for every task. – **2 marks**
3. **A house price is predicted using its area, number of rooms, and age. Which type of regression is suitable for this problem? Give a reason.** [U] / [CO2]
   Multiple Linear Regression. – **1 mark** The house price is a continuous dependent variable predicted using three independent variables: area, number of rooms, and age of the house. Multiple linear regression is suitable because it uses two or more independent variables to predict a continuous value - **1 mark**
4. **What is meant by linearly separable data?** [U] / [CO3]
   Linearly separable data refers to data in which different classes can be separated by a straight line in 2D or a hyperplane in higher dimensions. – **2 marks**
5. **Define a hyperplane in SVM.** [U] / [CO3]
   A hyperplane is a decision boundary that separates data points belonging to different classes. In SVM, the hyperplane is chosen to maximize the margin between the classes. – **2 marks**

[2x5 = 10]

Part B
Short Answer Questions
Answer any 5 questions
Each question carries 6 marks

6. **What is a Confusion Matrix? Explain its four components with a suitable example**. [K] / [CO1]
   A Confusion Matrix is a table used to evaluate the performance of a classification model in Machine Learning. It compares the actual class labels with the predicted class labels to determine how accurately the model classifies data. – **2 marks**
   Four Components of a Confusion Matrix – **2 marks**
   1. True Positive (TP): The model correctly predicts the positive class. 2. True Negative (TN):

The model correctly predicts the negative class. 3. False Positive (FP): The model incorrectly predicts the positive class when the actual class is negative. 4. False Negative (FN): The model incorrectly predicts the negative class when the actual class is positive.
Example - Consider a Machine Learning model that predicts whether students will pass or fail an examination. – **2 marks**

Actual / Predicted	Pass (Positive)	Fail (Negative)
Pass (Positive)	TP = 40	FN = 10
Fail (Negative)	FP = 5	TN = 45

Interpretation:
- TP = 40: 40 students who actually passed were correctly predicted as passed.
- TN = 45: 45 students who actually failed were correctly predicted as failed.
- FP = 5: 5 students who actually failed were incorrectly predicted as passed.
- FN = 10: 10 students who actually passed were incorrectly predicted as failed.

7. **A dataset has many features, but only some of them are useful. Explain how feature selection can help**. [A] / [CO2]
Feature selection is the process of selecting the most relevant and useful features from a dataset while removing irrelevant or redundant features. It helps improve the performance of a Machine Learning model. – **1 mark**
How feature selection helps - 1. Improves model accuracy: Removing irrelevant features can help the model make better predictions. 2. Reduces overfitting: It reduces the chance of the model learning unnecessary patterns from the training data. 3. Reduces computational time: Fewer features require less processing time and memory. 4. Simplifies the model: A model with fewer features is easier to understand and interpret. 5. Removes irrelevant data: It eliminates features that do not contribute significantly to the prediction. – **2 marks**
Example - Consider a dataset used to predict house prices. It contains the following features - Area of the house, - Number of rooms, - Age of the house, - House identification number, - Owner's name. The area, number of rooms, and age of the house may help predict the price. However, the house identification number and owner's name may not be useful. – **3 marks**
8. **Explain backpropagation and optimization in neural networks**. [U] / [CO3]
Backpropagation is an algorithm used to train neural networks by calculating the error between the predicted output and the actual output and propagating this error backward through the network. Steps involved in backpropagation:
1. Forward propagation: Input data passes through the network to produce a predicted output.
2. Error calculation: A loss function calculates the difference between the predicted output and the actual output.
3. Backward propagation: The error is propagated backward through the network using the chain rule to calculate gradients.
4. Gradient calculation: The gradients indicate how each weight and bias contributes to the error. – **3 marks**
Optimization is the process of adjusting the weights and biases of a neural network to minimize the loss function and improve prediction accuracy. Steps involved in optimization:
1. Calculate the gradients of the loss function. 2. Update the weights and biases using an optimization algorithm. 3. Repeat the process until the loss is minimized or a stopping condition is reached. Example: Gradient Descent is a commonly used optimization algorithm. – **3 marks**
9. **Demonstrate the working principle of BIRCH and mention its advantages.** [U] / [CO4]
BIRCH stands for Balanced Iterative Reducing and Clustering using Hierarchies. It is appropriate for very large datasets or for streaming data, because of its ability to find a good clustering solution with only a single scan of the data. Instead of storing all data points, BIRCH stores compact summaries called Clustering Features (CF). A CF contains the following. Number of data points (N), Linear sum of points (LS) and Squared sum of points (SS). – **4 marks. Advantages -** Fast and memory efficient. Suitable for very large datasets. Handles incremental data insertion. Reduces computational cost. – **2 marks**
10. **Explain agglomerative hierarchical clustering with its major steps.** [U] / [CO4]
Algorithm – Agglomerative Hierarchical Clustering (AGNES) – 6 marks

Input
D: a dataset containing n objects
Output: A set of k clusters
Method

1. Initialize i to n.
2. Find the distance between each pair of points.
3. Identify the two points (p, q) which are having the least distance between them.
4. Find the centroid of two points (p, q). Let them be ci and the distance be di.
5. Set i=i-1
6. If i>1 then go to step 3 else go to step 7.
7. Draw the dendrogram of the sequence of cluster formation.
8. Identify the sudden jump in the distance between any two clusters and identify the final set of clusters.
9. End

11 **Describe any three ethical considerations that should be addressed while developing and deploying machine learning systems.** [U] / [CO4]

1. Data Privacy and Security - Machine Learning systems often use large amounts of personal data. Developers must protect sensitive information and obtain appropriate consent before collecting and using data. Proper security measures should be implemented to prevent unauthorized access and data misuse. Example: A healthcare prediction system must protect patients' medical records and personal information. – **2 marks**

2. Fairness and Bias - Machine Learning models may produce biased results if they are trained using biased or unrepresentative datasets. Developers should use diverse and representative data and test models for unfair discrimination against individuals or groups. Example: A recruitment system should not discriminate against candidates based on gender or ethnicity. – **2 marks**

3. Transparency and Accountability - Machine Learning systems should be understandable to an appropriate extent, and their decisions should be explainable whenever possible. Developers and organizations must take responsibility for the system's outcomes and provide ways to review or correct incorrect decisions. Example: A bank should be able to explain why a loan application was rejected by its ML-based system. – **2 marks**

12 **Suppose a company wants to divide its customers into five groups based on purchasing behaviour. Explain how you would apply K-Means clustering to solve this problem.** [A] / [CO4]

Steps to Apply K-Means Clustering

1. Collect the Data
2. Preprocess the Data
3. Choose the Number of Clusters - Set the number of clusters to K = 5, as the company wants to divide its customers into five groups.
4. Initialize the Centroids - K-Means selects five initial centroids, which represent the centers of the five clusters.
5. Assign Customers to Clusters - Calculate the distance between each customer and the five centroids. Assign each customer to the cluster with the nearest centroid, usually using Euclidean distance.
6. Update the Centroids - Calculate the mean of the customers assigned to each cluster. These means become the new centroids.
7. Repeat the Process - Repeat the assignment and centroid-update steps until the centroids stop changing significantly or another stopping condition is reached.
8. Interpret the Clusters - Analyze the purchasing patterns of each group and assign meaningful labels based on the results. – **6 marks**

[6x5 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

13 **Explain Supervised Learning, Unsupervised Learning, Semi-Supervised Learning, and Reinforcement Learning with suitable examples.** [U] / [CO1]

1. Supervised Learning - Supervised Learning is a type of Machine Learning in which a model learns from labelled data containing both input features and their corresponding outputs. Working: - The model is trained using labelled data. - It learns the relationship between inputs and outputs. Classification and regression belong to supervised learning. It predicts the output for new, unseen data. Example: Predicting house prices using previous housing data containing house areas, numbers of rooms, and actual prices. Applications: Email spam detection, house price prediction, and disease diagnosis. – **4 marks**

2. Unsupervised Learning - Unsupervised Learning is a type of Machine Learning in which a model learns patterns and structures from unlabelled data without predefined output labels. The model receives unlabelled data. It identifies similarities, patterns, or relationships. It groups similar data points or discovers hidden structures. Clustering comes under unsupervised learning. Example: A company uses K-Means clustering to divide customers into five groups based on their purchasing behaviour. Applications: Customer segmentation, anomaly detection, and grouping similar documents. – **4 marks**

3. Semi-Supervised Learning - Semi-Supervised Learning is a type of Machine Learning that uses a small amount of labelled data along with a large amount of unlabelled data for training. A small portion of the data is labelled. A large portion remains unlabelled. The model uses both types of data to improve its learning and prediction performance. Example: In image classification, a model is trained using 500 labelled images and 5,000 unlabelled images to classify objects. Applications: Medical image analysis, speech recognition, and image classification. – **3 marks**

4. Reinforcement Learning - Reinforcement Learning is a type of Machine Learning in which an agent learns to make decisions by interacting with an environment and receiving rewards or penalties. The agent observes the environment. It takes an action. The environment provides a reward or penalty. The agent learns to choose actions that maximize its cumulative reward over time. Example: A robot learns to navigate a room by receiving rewards for reaching its destination and penalties for hitting obstacles. Applications: Robotics, game playing, and autonomous navigation. – **4 marks**

14 **Explain the steps involved in constructing a Decision Tree using a suitable example. Explain the concepts of Entropy and Information Gain and describe how they are used to select the best splitting attribute. Discuss the advantages, disadvantages, and major issues in Decision Tree learning.** [An] / [CO2]

1. Steps Involved in Constructing a Decision Tree – **5 marks**

Step 1: Collect the training data

Step 2: Select the best splitting attribute - Choose the attribute that best separates the training examples into different classes. Entropy and Information Gain are commonly used for this purpose.

Step 3: Create the root node - Select the best splitting attribute as the root node of the tree.

Step 4: Divide the dataset - Split the dataset into subsets according to the possible values of the selected attribute.

Step 5: Repeat the process - For each subset, select the next best attribute and repeat the splitting process until the stopping condition is reached.

Step 6: Create leaf nodes - Create a leaf node when all examples in a subset belong to the same class or when another stopping condition is satisfied.

Step 7: Prune the tree- Remove unnecessary branches that may cause overfitting and reduce the model's ability to generalize to unseen data.

2. Entropy - Entropy is a measure of impurity or uncertainty in a dataset. It indicates

how mixed the class labels are in a particular dataset. 3. Information Gain - Information Gain (IG) measures the reduction in entropy achieved by splitting a dataset using a particular attribute. – **2 marks**

Advantages of Decision Trees – **2 marks**

- ✓ 1. It does not require any domain knowledge.
- ✓ It is easy to comprehend.
- ✓ The learning and classification steps of a decision tree are simple and fast.

Disadvantages of Decision Trees – **2 marks**

1. Overfitting: A very deep tree may learn noise and unnecessary details from the training data. 2. Instability: Small changes in the training data can produce a significantly different tree. 4. Greedy approach: The algorithm selects the best split at each step, but this does not guarantee a globally optimal tree. 5. Complexity: Large trees can become difficult to interpret.

Major Issues in Decision Tree Learning – **4 marks**

1. Overfitting - A decision tree may learn the training data too well, including noise and outliers. It means it performs very well on training data but performs poorly on unseen test data.

2. Underfitting - If the tree is too small or simple, it cannot capture patterns in the data. Hence poor accuracy on both training and testing data.

3. Handling continuous attributes - Decision trees naturally work well with categorical data, but continuous data requires selecting split points.

4. Data fragmentation - As the tree grows deeper, fewer samples remain in each node. The result is decisions become unreliable and there is increased chance of overfitting.

5. Bias toward attributes with many values - Algorithms using information gain may prefer attributes with many unique values. Example: Student ID, Phone Number. These attributes may not actually help prediction. .

6. Noise and missing values - Incorrect or missing data can affect splitting decisions. It may result in wrong classification, reduced accuracy.

7. Instability - Small changes in training data can create a completely different tree. For example, removing a few records may change the root node.

8. Complexity of Large Trees - Large trees are difficult to interpret and hard to visualize. One advantage of decision trees is interpretability, which decreases with very deep trees.

9. Greedy nature of algorithm - Decision trees use a greedy approach in which best split is chosen at each step. It does not guarantee a globally optimal tree.

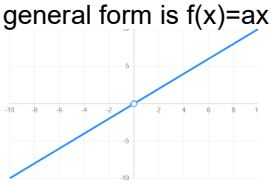
10. Class imbalance problem - If one class dominates the dataset, the tree may become biased toward that class. The solutions are resampling techniques, balanced datasets and class weights.

15 **What is activation function? Explain any 4 activation function in detail with its mathematical formula and working.** [An] / [CO3]

[15x2 = 30]

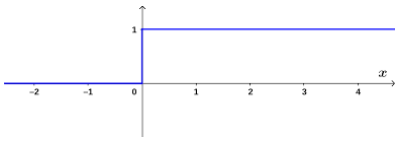
Activation functions are mathematical functions used in machine learning and neural networks to determine the output of a neuron. They introduce nonlinearity into the network, allowing neural networks to learn complex patterns. Without activation functions, a neural network behaves like a simple linear model. – **1 mark**. The commonly used activation functions are linear, step, sigmoid, tanh, and rectified linear unit (ReLu) functions as given below. (Any 4 – **3.5 marks each**)

Linear function – A linear function is a mathematical function in which the variable has a power of 1. The general form is f(x)=ax



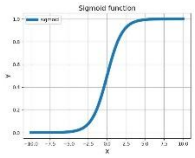
Step function – A step function is a function whose value remains constant within certain intervals and changes suddenly at specific points. It looks like a series of “steps” instead of a smooth curve. The general form is given below.

$$f(x) = \begin{cases} 0 & \text{for} \quad x < 0 \\ 1 & \text{for} \quad x \geq 0 \end{cases}$$



Logistic (Sigmoid) function – A logistic function or Sigmoid function is a non-linear function commonly used in machine learning and neural networks. The mathematical form is given below.

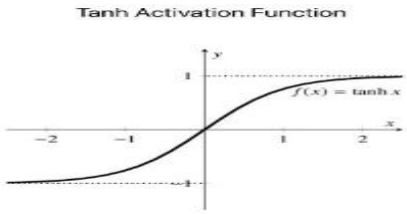
$$f(x) = \frac{1}{1+e^{-x}}$$



It produces an S-shaped curve. It is used to convert values into probabilities. The graph starts near 0, increases gradually, passes through (0,0.5) and approaches 1 asymptotically. It has applications in binary classification, neural network activation functions, logistic regression and probability estimation.

Tanh function – A Tanh Function (Hyperbolic Tangent Function) is an activation function widely used in machine learning and neural networks. Its mathematical form is given below.

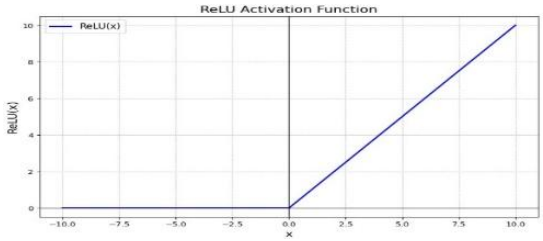
$$f(x) = \tanh(x) = \frac{2}{1+e^{-2x}} - 1$$



The graph starts near −1, passes through (0,0) and gradually approaches 1. It has applications in hidden layers in neural networks, deep learning models and signal processing.

Rectified linear unit (ReLu) function – A Rectified Linear Unit (ReLU) is one of the most commonly used activation functions in machine learning and deep learning. Its mathematical form is given below.

$$f(x) = \begin{cases} 0 & \text{for } x < 0 \\ x & \text{for } x \geq 0 \end{cases}$$



The shape of the graph is flat line for negative x and straight line with slope 1 for positive x. It is simple and computationally efficient. It is a non-linear activation function. It is easy to implement and has faster training in deep neural networks.

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